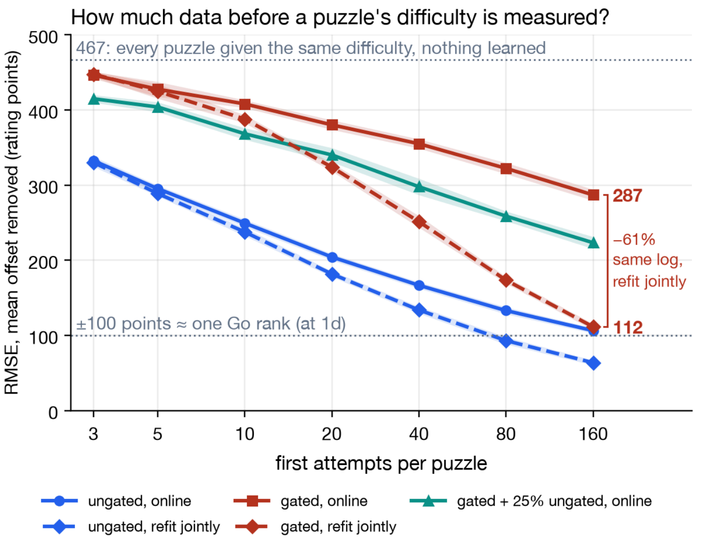
Measuring Go puzzle difficulty

How much attempt data does it take to measure a puzzle's difficulty, instead of assigning it by hand?

What I did. Plant 300 puzzles and 3,000 players with known difficulties and skills. Simulate their first-attempt logs under three exposure patterns; the one modelling Go Magic is skill-tree gating, where a player meets only puzzles within 300 points of their own level. Fit the ungated and gated logs twice with the truth hidden: online, one attempt at a time as a live system runs Glicko-2, and jointly, one Rasch fit of every rating at once. Score both against the planted difficulties. Scale: **±100 points ≈ one Go rank** (EGF convention; measured, a rank is 96 points at 1d and 44 at 10k); assigning every puzzle the same difficulty scores 467 points of error.

About 160 first attempts measure a puzzle's difficulty to roughly one rank, but on skill-tree data only if the log is fitted *jointly*. Replaying the same log one attempt at a time, the standard online way, leaves almost three ranks of error: 287 points against 112, a 61% cut for nothing but compute.

So backfill the existing log jointly. The trap is replaying it online, reading 287 points of error and calling the data inadequate.



Three exposure patterns, two estimators. Colour is the data: ungated random pairing against skill-tree gating, a ±300-point band standing in for Go Magic's row-by-row tree (which Gold and Magic members can switch off, so part of the real log is ungated); green, fitted online only, adds 25% ungated traffic. Line style is the estimator: solid online, dashed the same log refit jointly. Dotted lines: one-rank accuracy and the 467 ceiling. Bands: 95% intervals over ten simulated worlds.

Reproduce. `./src/recovery.py --puzzles 300 --reps 10` draws every curve here; the other numbers trace to `docs/RUNNING.md` and `METHOD.md` sections 1 and 7; the estimator is checked against Glickman's worked example. Go Magic facts are from the committed snapshot of its public skill-tree page (16 Aug 2026); nothing private.

THREE RESULTS

Gating itself costs accuracy

Under gating no player's attempts span an easy puzzle and a hard one: against ungated pairing, **1.6×** the error at 10 first attempts per puzzle, **2.7×** at 160, and a coarser check holds about 2.8× out to 1,280. Neither more traffic nor a better estimator closes the gap to ungated data: refit jointly, **1.75×** remains.

The damage is to the spacing, not the ordering

At 160 attempts the gated online fit ranks puzzles well but compresses the scale to under half its width: it knows *which* puzzle is harder, not *by how much*. The joint fit repairs most of it.

Uneven traffic is a second, separate cost

Everybody attempts the first row; few reach the last. At equal volume an assumed 50:1 funnel costs **30–50** points and drops the ordering, the one thing gating had left intact, from 0.94 to **0.78**.

at 160 first attempts	RMSE	compression	ordering ρ
ungated, online	107	1.20	0.986
ungated, joint	64	1.09	0.994
gated, online	287	2.36	0.945
gated, joint	112	1.30	0.996

RMSE in rating points, mean offset removed. Compression: the slope of planted on fitted difficulty; 1.0 is right, 2.36 means the fitted spread is 2.36× too narrow. ρ: rank correlation. Ten-world means; each pair of rows fits the identical log.

WHAT TO DO ABOUT IT

Backfill jointly, serve online

Fit the history jointly for the labels, with one skill per player per time window: the joint fit otherwise holds a learner's skill fixed. Keep online Glicko-2 for the live path, where one-at-a-time updating is right.

Ship labels per puzzle, not per catalogue

Publish a measured label where enough attempts exist and keep the hand label until then, decided by count, not by Glicko's own error bar (RD), which under gating understates its error **3.7×**.

Anchor the scale with an ungated test

An ungated test across the ranks is the standard repair for compression: 25% of traffic through one buys the online fit back **57** points at 40 attempts and half the compression. Go Tests may already be one.

What this does not claim. Not that any hand label is wrong; that needs the private attempt log, and none is used here. It answers the question before that one: measured from attempts, how much data would a difficulty need to mean anything? Simulation can settle that for the estimator and the data's shape; real attempts fit its one-trait logistic model less well, so read the counts here as a floor.

The next step is small: fit one month of the real log both ways, first attempts only (the try the coin rule singles out). If the table holds, the online labels come out about half as wide as the joint ones, and attempts per puzzle place every label on the red curves above.